

Estimating and Solving with Decimals

Answer Key

Grades 4–5 number and operations in base ten

Quick Answers

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|---------------------------|----------------------|
| 1. 6 | 7. 30 |
| 2. 80 | 8. 1.8 |
| 3. 24.85 | 9. 13.8, — |
| 4. 14.4 | 10. 10.6, 0.6 |
| 5. 0.85, 0.89, 0.9, 0.905 | 11. 4.2, 7 |
| 6. 0.85, 1.65 | 12. 34, 34.2, —, 5.8 |
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Curriculum

Level: Grades 4–5 number and operations in base ten

4.NBT.A.1–A.3 / 5.NBT.A.1–A.2 – *problem 5*

5.NBT.B.7 / 6.NS.B.3 – *problems 1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 12*

Difficulty 1–5 of 5 across 19 tagged checks.

Common wrong answers

- 3. *If they got 31.25: subtracted only the first purchase*
 - 4. *If they got 1.44: put the decimal point one place too far left*
 - 8. *If they got 28.8: multiplied by the number of pieces instead of dividing*
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- 1.** Notebook \$3.89, pen \$2.15. Estimate the total to the nearest whole dollar.

Solution: Round *first*, then add. 3.89 is between 3 and 4 and its tenths digit is 8, so it rounds up to 4; 2.15 has tenths digit 1, so it rounds down to 2. Then $4 + 2 = 6$. (The exact total is \$6.04, so the estimate is a good one.) 6

- 2.** 48.6 km one week, 27.3 km the next. Estimate the total to the nearest ten kilometres.

Solution: Round each distance to the nearest ten before adding. 48.6 is nearer to 50 than to 40; 27.3 is nearer to 30 than to 20. Then $50 + 30 = 80$. 80 km

- 3.** \$50 in the account; \$18.75 and \$6.40 spent. How much is left?

Solution: Two steps. First add what was spent, lining up the decimal points: $18.75 + 6.40 = 25.15$. Then subtract from the starting amount, writing the 50 as 50.00 so the columns line up: $50.00 - 25.15 = 24.85$. Check by adding back: $25.15 + 24.85 = 50.00$ ✓ 24.85

4. 6 bags of rice at 2.4 kg each. Exact total mass?

Solution: $6 \times 24 = 144$, and 2.4 has one digit after the decimal point, so the answer has one digit after the decimal point too: 14.4. Estimate as a check: $6 \times 2 = 12$, so an answer near 12 is sensible — 1.44 would have been far too small.

14.4 kg

5. Order 0.9, 0.85, 0.905 and 0.89 from shortest to longest.

Solution: Give every number the same number of decimal places first, by writing zeros on the end (which does not change a decimal's value): 0.900, 0.850, 0.905, 0.890. Now compare them as though they were whole numbers of thousandths: $850 < 890 < 900 < 905$.

$0.85 < 0.89 < 0.9 < 0.905$

6. Recipe with 1.25 L milk and 0.4 L cream. (a) How much more milk? (b) How much altogether?

Solution (a): Write 0.4 as 0.40 so the columns line up, then subtract: $1.25 - 0.40 = 0.85$.

0.85 L

Solution (b): Add the same two amounts: $1.25 + 0.40 = 1.65$.

1.65 L

7. 6 folders at \$4.85 each. Estimate the total to the nearest whole dollar.

Solution: Round the price first: 4.85 is nearer to 5 than to 4. Then $6 \times 5 = 30$. (The exact cost is \$29.10, so the estimate is about 90 cents high — which is what you would expect, since every folder was rounded up.)

30

8. A 7.2 m ribbon cut into 4 equal pieces. How long is each?

Solution: Sharing equally means dividing: $7.2 \div 4$. Since $72 \div 4 = 18$ and 7.2 has one decimal place, each piece is 1.8 m. Check by multiplying back: $4 \times 1.8 = 7.2$ ✓ — and note that each piece must be *shorter* than the whole ribbon, so an answer bigger than 7.2 cannot be right.

1.8 m

9. Sam wrote $4.6 \times 3 = 1.38$. (a) Find the exact product. (b) Compare it with Sam's answer. (c) What went wrong?

Solution (a): $46 \times 3 = 138$, and 4.6 has one decimal place, so the product has one decimal place: 13.8.

13.8

Solution (b): 13.8 is much bigger than 1.38.

$13.8 > 1.38$

Solution (c): Sam put his decimal point two places from the right instead of one, giving hundredths where the answer should have tenths — his answer is a tenth of the true product. An estimate catches this immediately: 4.6 is nearly 5, and $5 \times 3 = 15$, so the answer has to be near 15, not near 1.

10. A hike of 3.75 km, 2.8 km and 4.05 km. (a) Total distance. (b) How far past a 10 km target?

Solution (a): Write all three with two decimal places and add: $3.75 + 2.80 + 4.05 = 10.60$.

10.6 km

Solution (b): Subtract the target from the total: $10.60 - 10.00 = 0.60$.

0.6 km

11. A tap delivers 0.35 L a second. (a) Water after 12 seconds. (b) How many 0.6 L jugs does that fill?

Solution (a): $35 \times 12 = 420$, and 0.35 has two decimal places, so the product has two: 4.20 L, which is 4.2 L.

4.2 L

Solution (b): Dividing by a decimal is easier if both numbers are made ten times bigger — that does not change the answer: $4.2 \div 0.6 = 42 \div 6 = 7$. Check: $7 \times 0.6 = 4.2$ ✓

7

12. 4 packs of paper at \$6.75 and 3 markers at \$2.40, from \$40. (a) Estimate. (b) Exact total. (c) Compare. (d) Money left.

Solution (a): Round each price first: 6.75 rounds to 7 and 2.40 rounds to 2. Then $4 \times 7 = 28$ and $3 \times 2 = 6$, giving $28 + 6 = 34$.

34

Solution (b): Exactly: $4 \times 6.75 = 27.00$ and $3 \times 2.40 = 7.20$, so the total is $27.00 + 7.20 = 34.20$.

34.2

Solution (c): The estimate came out slightly below the exact total, because the markers were rounded down by 40 cents each while the paper was only rounded up by 25 cents each.

$34 < 34.2$

Solution (d): Subtract the exact total from what the class had: $40.00 - 34.20 = 5.80$. Check: $34.20 + 5.80 = 40.00$ ✓

5.8